PENETRATION TESTING ASSESSMENT REPORT

Romando Wright

Springboard Cybersecurity Career Track

Authorized Lab-Based Security Assessment

Client:

Artemis(Simulated Environment)

January 10, 2026

Abstract

This report presents the findings of a security assessment conducted on the Artemis environment using a controlled penetration-testing approach. The assessment focused on identifying potential vulnerabilities, evaluating realistic threat scenarios, and providing actionable recommendations to reduce risk.

Key findings included exposed remote access services and the use of self-signed encryption certificates, which could allow unauthorized access, interception of sensitive information, and potential expansion of compromise within the environment. While no active exploitation was performed, the findings reflect realistic attack paths that modern threat actors commonly use.

The report provides clear, practical recommendations, including limiting remote access, using trusted encryption certificates, implementing multi-factor authentication, and monitoring system activity. Addressing these issues strengthens operational resilience, protects sensitive information, and helps

maintain stakeholder trust.

Acknowledgements

The completion of this assessment and report would not have been possible without the support and cooperation of the Artemis team. Their guidance, access to the lab environment, and timely responses enabled a thorough evaluation of the security posture. Special thanks are extended to all team members who assisted in facilitating the assessment process and ensuring the findings could be accurately documented for management review.

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Executive Summary (Management Report)

Executive Summary

This assessment of the Artemis environment's security posture employed a controlled penetration-testing approach to identify vulnerabilities that could pose risks to the organization. Significant weaknesses were found in externally accessible remote access services and cryptographic trust configurations. Although active exploitation was not performed, these vulnerabilities reflect potential attack pathways used by modern threat actors. If unaddressed, they may lead to unauthorized access, interception of sensitive information, and lateral movement, potentially resulting in service disruptions, heightened incident response costs, regulatory issues, and long-term reputational harm.

Key Business Risks Identified

Risk Area	Description	Livelihood	Business Impact	Overall Risk
Remote Access Exposure	Remote Desktop Protocol (RDP) services are accessible beyond trusted network boundaries, increasing the risk of unauthorized access.	Medium	High	Medium
Cryptographic Weakness	Use of self-signed SSL certificates reduces trust assurance and increases exposure to interception and impersonation attacks.	Medium	Medium	Medium
Lateral Movement Potential	An initial compromise could allow attackers to expand control across internal systems and services.	Low-Medium	High	Medium

Business Impact Overview

From a business perspective, gaps in remote access controls increase the chances of unauthorized access and interruptions to normal operations. If these issues were taken advantage of, the organization could experience service downtime, higher investigation and recovery costs, possible compliance concerns, and loss of trust from customers and partners.

Addressing these issues early will greatly reduce risk and help strengthen the organization's ability to operate safely and reliably.

Risk Visualization

Figure 1: Risk Likelihood vs Business Impact Matrix summarizing identified security findings

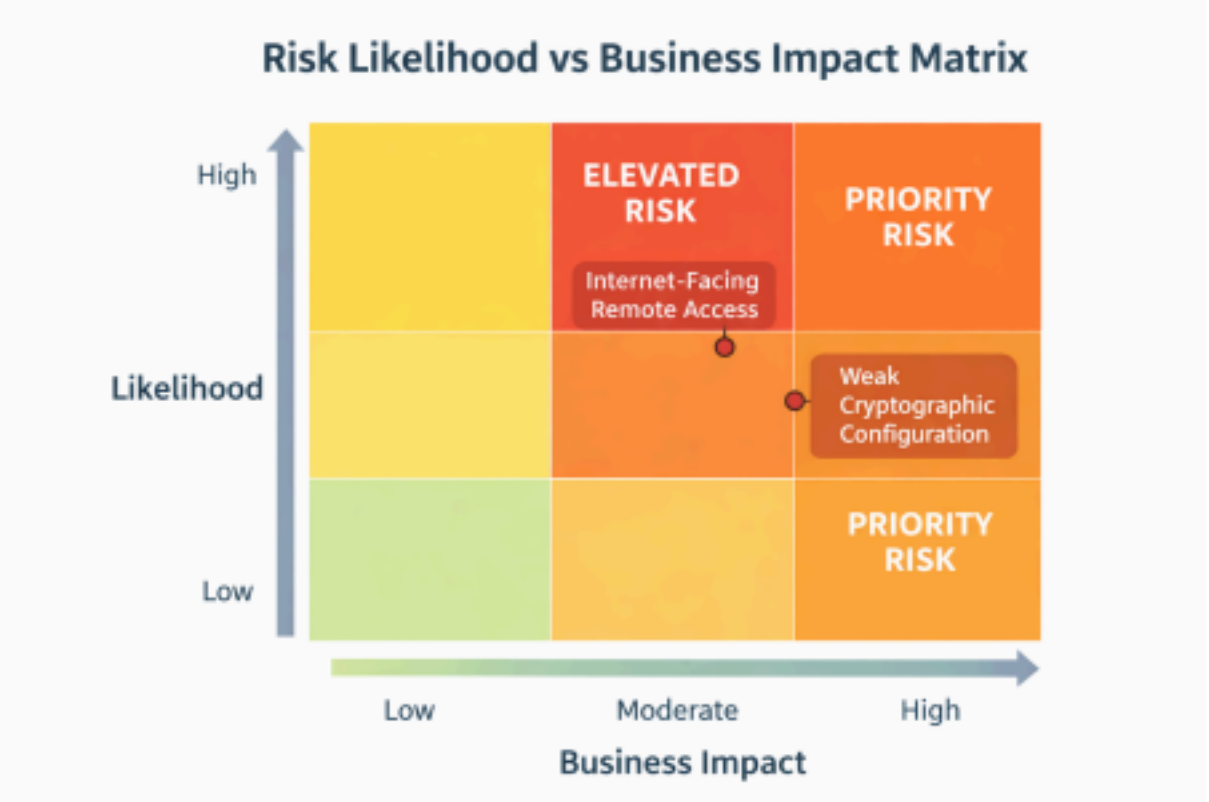

Figure 1 illustrates that externally accessible remote access services represent an elevated risk when combined with moderate-to-high business impact.

Executive Recommendations

To strengthen the organization's security posture and reduce business risk, the following actions are recommended:

- Replace self-signed security certificates with certificates issued by trusted providers to improve trust and data protection.

- Limit remote access so it is only available to approved users and locations through improved network controls.
- Require additional identity verification for administrative access to prevent unauthorized use of critical systems.
- Perform regular system reviews and security assessments to ensure controls remain effective as the environment evolves.

Detailed Technical Report

2.1 Scope of Work

This assessment was conducted on a single system within a controlled VirtualBox lab environment. The work focused on identifying potential weaknesses and evaluating possible risks. No systems were disrupted, and no harmful actions were taken during the assessment.

The activities were strictly limited to observation, analysis, and documentation of security concerns to ensure the environment remained safe and unaffected.

2.2 Project Objectives

The objectives of this assessment were to:

- Identify systems and services that may be unnecessarily exposed.
- Review configurations for weaknesses that could increase risk.
- Evaluate realistic scenarios in which these weaknesses could be exploited.
- Provide practical recommendations to reduce risk and improve security.

2.3 Assumptions

This assessment was based on the following assumptions:

- The lab environment closely reflects a real-world production setup.
- All assessment activities were fully authorized and conducted in a safe, non-disruptive manner.
- The findings represent realistic security exposures commonly seen in organizational environments.

2.4 Timeline

Phase 1	Reconnaissance
Phase 2	Target Identification
Phase 3	Vulnerability Identification
Phase 4	Threat Assessment
Phase 5	Reporting

2.5 Reconnaissance Activities

During the initial phase of the assessment, the team identified active systems and services that were accessible. This process revealed that remote access functionality was available from outside the trusted environment, creating a primary point of exposure.

This information formed the basis for the risk analysis and subsequent recommendations.

2.6 Identified Vulnerabilities

During the assessment, we found a few key weaknesses that could put the organization at risk:

- Remote Access Exposure: Some systems can be accessed from outside the company's trusted network. This means someone with the right tools could try to access the system without permission.

- Weak Encryption Certificates: The system uses self-signed certificates for encrypted communications. These are like "homemade locks" rather than professionally trusted ones, which makes it easier for attackers to intercept sensitive information.

These findings were checked carefully with multiple tools to make sure they were accurate and not false alarms.

2.7 Threat Analysis

If the vulnerabilities above were exploited, the potential consequences include:

- Interception of sensitive information: Communications could be "listened to" by attackers if encryption is weak.

- Unauthorized system access: Attackers could gain access to systems they shouldn't be able to reach.

- Spread of attacks: Once inside, attackers could move to other systems, potentially causing wider damage.

- Higher privileges for attackers: If administrative access is compromised, attackers could take full control of certain systems.

These are realistic threats and are similar to methods used by attackers in real-world situations. Addressing these issues now helps prevent costly

2.8 Recommendations

To reduce risk and protect the organization, the following steps are recommended:

1. Use trusted certificates: Replace self-signed certificates with professionally issued ones to ensure communications are secure.
2. Limit remote access: Only allow remote access for approved users and from trusted locations.
3. Enable extra security for remote access: Use Network Level Authentication (NLA) to add an extra layer of protection.
4. Add multi-factor authentication: Require a second form of verification, like a code sent to a phone, for all administrative access.
5. Monitor activity: Keep an eye on remote access logs to detect unusual or suspicious behavior early.

These actions are practical steps that greatly reduce the risk of security breaches and protect the organization's operations.

2.9 Conclusion

This assessment shows that misconfigured remote access and weak encryption can create serious risks for the organization. Even though no systems were harmed during this assessment, these vulnerabilities represent realistic paths that attackers could take.

By following the recommended actions, the organization can:

- Protect sensitive information
- Prevent unauthorized access
- Reduce potential downtime and recovery costs
- Maintain trust with customers and stakeholders

Taking proactive steps now strengthens the organization's ability to operate safely and reliably.

References

Microsoft Corporation. (n.d.). *Remote Desktop Services overview*. Microsoft Learn. https://learn.microsoft.com/en-us/windows-server/remote/remote-desktop-services/

Microsoft Corporation. (n.d.). *Understanding Remote Desktop Protocol (RDP)*. Microsoft Learn. https://learn.microsoft.com/en-us/troubleshoot/windows-server/remote/understanding-remote-desktop-protocol

OWASP. (n.d.). *Transport Layer Security cheat sheet*. OWASP Cheat Sheet Series. https://cheatsheetseries.owasp.org/cheatsheets/Transport_Layer_Security_Cheat_Sheet.html

Cybersecurity & Infrastructure Security Agency. (n.d.). *Multifactor authentication*. https://www.cisa.gov/topics/cybersecurity-best-practices/multifactor-authentication

SentinelOne, Inc. (n.d.). *Remote Desktop Protocol (RDP)*. SentinelOne. https://www.sentinelone.com/cybersecurity-101/identity-security/remote-desktop-protocol/

Wikimedia Foundation. (2025, December 1). *Remote Desktop Protocol*. Wikipedia. https://en.wikipedia.org/wiki/Remote_Desktop_Protocol

CheapSSLShop. (n.d.). *What is SSL/TLS certificate and how does it work?* https://www.cheapsslshop.com/blog/what-is-an-ssl-tls-certificate-and-how-does-it-work

Cybersecurity & Infrastructure Security Agency. (n.d.). *Multi-factor authentication (MFA) toolkit*. https://www.cisa.gov/resources-tools/resources/multi-factor-authentication-mfa-toolkit